

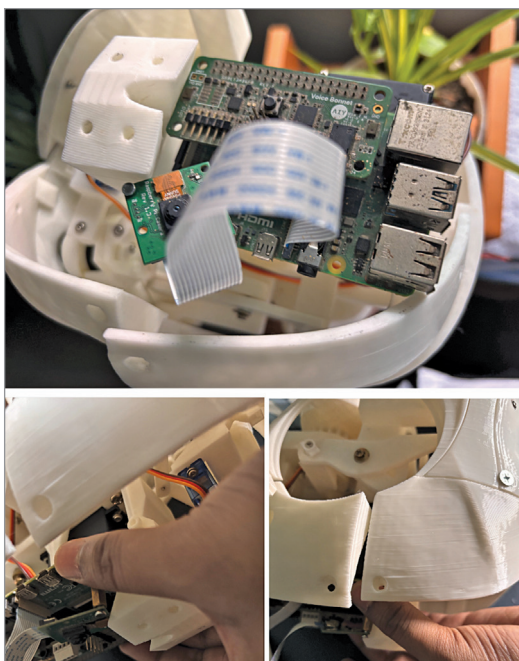


Fig. 4: Installing the robot's computing unit inside the head

used to drive the servos. Alternatively, servos can be driven directly from the Raspberry Pi GPIO pins with a 5V supply. In that case, update the PWM signal pin in the code accordingly.

Finally, connect the camera via the CSI cable and port.

Software

To start coding, assume that the Raspberry Pi includes an SD card preloaded with the Raspbian OS. If not, install and set up the OS by referring to the official Raspberry Pi website.

The Python IDE is generally pre-installed. If not, install it separately.

Next, install MediaPipe and the required Python modules for image processing and servo control. Fig. 6 shows the MediaPipe installation process.

Installing MediaPipe. Open the terminal and run the following command: `sudo pip3 install mediapipe`

Verifying the installation. Start Python and import MediaPipe by typing: `import mediapipe as mp`

If no errors appear, it means MediaPipe has been installed correctly.

Installing servo control library. The gpiozero library is used to control the servo motors. Install it using the following command: `sudo pip3 install gpiozero`

Installing OpenCV. Open the terminal and run the following command: `sudo pip3 install opencv-python`

Downloading the object detection model. MediaPipe supports multiple object detection models. One efficient option for Raspberry Pi is efficientdet_lite0.tflite, which offers a balance between speed and accuracy.

Download the model from the official MediaPipe page: https://ai.google.dev/edge/mediapipe/solutions/vision/object_detector

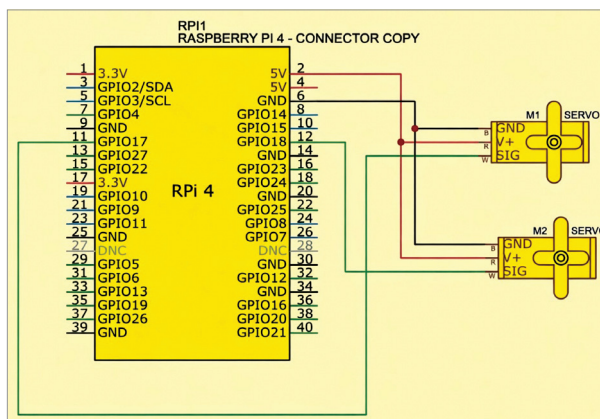


Fig. 5: Circuit diagram

```
requirement already satisfied: matplotlib in /usr/local/lib/python3.7/dist-packages (fr
requirement already satisfied: wheel in /usr/local/lib/python3.7/dist-packages (from me
collecting absl-py (from mediapipe-rpi4)
Downloading https://files.pythonhosted.org/packages/a2/ad/e8d3c824784ff121c83cc831f94
py-2.1.0-py3-none-any.whl (133kB)
100% |#####| 143kB 1.3MB/s
requirement already satisfied: protobuf>=3.11.4 in /usr/local/lib/python3.7/dist-packag
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requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.7/dist-packa
requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.7/dist-pa

requirement already satisfied: pillow>=6.2.0 in /usr/local/lib/python3.7/dist-packages
installing collected packages: absl-py, mediapipe-rpi4
Successfully installed absl-py-2.1.0 mediapipe-rpi4-0.0.8
You are using pip version 9.0.1, however version 25.3 is available.
You should consider upgrading via the 'pip install --upgrade pip' command.
@raspberrypi:~$ python3
Python 3.7.3 (default, Oct 31 2022, 14:04:08)
GCC 8.3.0 on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>> import mediapipe as mp
```

Fig. 6: MediaPipe installation on Raspberry Pi

Place the downloaded .tflite file in the same directory as your Python code. If the model file is in the same folder, you only need to specify the filename in the code.

Next, two different programs need to be prepared. The first program is for face tracking, which detects more than 100 facial landmark points in real time and moves the servo motors to track the face. The second program is for object tracking.

Face tracking code. Face tracking is implemented using Python libraries such as gpiozero, mediapipe, and OpenCV. The program captures camera frames in real time, processes them, and passes them to MediaPipe. MediaPipe returns coordinates of facial landmark points, and the servo motors move accordingly to track the face.

To connect the eye and head servo motors to the X and Y coordinates of the detected point, the

code needs to be adjusted accordingly. Fig. 7 shows the code snippet for face tracking, including configuring the servo control pins and importing the AI-based image-processing modules.

Object tracking code. In the object tracking code, import the MediaPipe library and specify the object detection model name and path that you downloaded earlier. If the model file is kept in the same folder as the Python program, you do not need to specify the full path; simply mention the model filename in the code. Refer to